Leveraging Supervised Machine Learning Techniques in Detecting Anomalies in Food Commodity Prices: Insights for Kenya's Market Stability

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Declaration and Approval

I declare that this work has not been previously submitted and approved for the award of a degree by this or any other university. To the best of my knowledge and belief, the Thesis contains no material previously published or written by another person except where due reference is made in the research itself.

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Abstract

Price volatility in Kenya's food commodity markets has a significant impact on economic stability and food security, posing substantial risks such as financial losses for farmers and traders, as well as increased vulnerability to food shortages for communities. The current manual processes employed to detect anomalies in food commodity pricing are inefficient, slow, and computationally demanding, resulting in delayed and inadequate responses to market disruptions. This presents a critical challenge, as timely detection and intervention are essential for maintaining market stability and ensuring food availability. This research proposes the application of supervised machine learning techniques to systematically identify pricing anomalies. The study aims to evaluate and select the most effective and suitable model for anomaly detection. Leveraging historical commodity pricing data, the study intends to demonstrate how these machine learning approaches can provide a robust alternative to manual analysis. The expected outcomes include the development of a scalable, efficient, and user-friendly cloud-based system capable of real-time anomaly detection and identification. This system will facilitate continuous learning from emerging market data, enabling prompt identification of unusual pricing trends. Consequently, stakeholders will gain immediate and actionable insights, empowering proactive decision-making and timely interventions. The ultimate goal is to substantially enhance Kenya's food market stability and strengthen food security by providing policymakers, traders, and farmers with precise market intelligence.

Keywords: *price volatility, anomaly detection, machine learning, food security, economic stability, supervised learning, Kenya*

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List of Abbreviations

RNN Recurrent Neural Networks

NDVI Normalised Difference Vegetation Index

SPI Standardised Precipitation Index

ALPS Alert for Price Spikes

CPI Consumer Price index

FEWS NET Famine Early Warning Systems Network

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CHAPTER 1: INTRODUCTION

1.1 Background Information

Food price stability is crucial for economic and social well-being, particularly in developing countries like Kenya, where a significant portion of household income is spent on food (Yusuf and Adelabu, 2024). According to Gbashi and Njobeh (2024), anomalies in food prices can have consequences that not only affect food security but also inflation and the overall stability of the food commodity market. Thus, food prices are a critical component in stabilising the market, yet they are subject to significant and often unanticipated fluctuations due to diverse economic, political and environmental changes. These irregularities in price, referred to as anomalies, can potentially destabilise markets, strain household budgets, and exacerbate food insecurity. This, therefore, implies that there is a need for timely identification of price anomalies to inform policy decisions and enable timely interventions to support the market.

A variety of factors contribute to anomalies in food prices in developing countries, including Kenya. A study by Brown and Kshirsagar (2015) highlights international shocks and weather as key factors causing price fluctuations. Weather, as a central component, determines seasonal patterns that affect harvest cycles, which are critical for establishing predictable changes in supply and demand. On the contrary, unexpected disruptions such as drought, floods, geopolitical instability and supply chain breakdowns lead to sudden and unpredictable drops or spikes in food prices (Awange et al., 2007). These price anomalies may have extreme implications that affect the ability of households to access affordable food and threaten the sustainability of agricultural producers.

However, as observed by Ilaha (2022), the traditional methods of monitoring and predicting fluctuations in food prices occasionally fail to capture complex market dynamics and identify subtle anomalies that may indicate emerging trends or potential crises. The author further indicates that relying on traditional statistical methods such as analytical indicators, factor modelling, comparative and grouping techniques, as well as manual monitoring, which are common in foundational economic analysis, is labour-intensive and time-consuming. Equally, as postulated by Pitchaimuthu and Holla (2024), while these methods can provide useful insights, they are often limited in their ability to process large datasets or capture complex patterns, especially when multiple variables interact. Thus, coupled with the growing availability of detailed food price data, there is an urgent need for more advanced and scalable tools that can quickly help detect and interpretation of anomalies.

Therefore, this study proposes machine learning techniques as a solution to addressing these challenges. In contrast with traditional methods, machine learning models are suitably placed because of their capability to process large volumes of data and identify hidden patterns that can reveal price anomalies (Korir et al., 2020). These models can handle non-linear relationships and complex interactions between variables, making them suitable for the intricate dynamics common in food price markets. For instance, deep learning models like recurrent neural networks (RNNs) and long short-term memory (LSTM) networks have demonstrated particular advantages in capturing temporal dependencies in time series data (Santoro et al., 2024). Similarly, studies have demonstrated that machine learning algorithms like eXtreme gradient boosting (XGBoost) have often performed better than the RNN-LSTM model for predicting accuracy compared to other shallower algorithms using stationary data.

1.2 Problem Statement

Kenya's food commodity markets are characterised by frequent and uneven price fluctuations, influenced by a range of factors including seasonal production cycles, climate variability, transport constraints, policy shifts and global commodity market dynamics (Nyoro et al., 2004; FAO, IFAD, UNICEF, WFP and WHO, 2022). While some of these price shifts reflect genuine market shocks, such as droughts, flooding or disruptions in supply chains, others are a result of anomalies in data collection, reporting errors or temporary localised events (Gbadegesin et al., 2024).

However, the existing monitoring systems typically depend on manual data inspection or simple rule-based thresholds applied to aggregated datasets. These approaches often fail to capture subtle yet quite important deviations, and they are also slow in generating alerts and likely to often produce false positives and false negatives (Gustafson, 2013). This undermines timely decision-making with negative impacts for policymakers, traders and humanitarian agencies to implement targeted interventions for stabilising prices and maintaining market resilience. The study proposes supervised machine learning techniques to support the development of the capacity to learn complex, non-linear patterns in historical price data, accounting for seasonality, spatial heterogeneity and inter-commodity relationships more effectively than conventional statistical methods (Jordan and Mitchell, 2015; Aggarwal, 2017).

1.3 Research Objectives

1.3.1 General Objectives

This study aims to develop a supervised machine learning framework for detecting anomalies in food commodity prices to strengthen Kenya's market monitoring and stability.

1.3.2 Specific Objectives

The following specific objectives will be adopted in implementing this study;

i) To develop a data engineering process that captures seasonality, spatial variation and inter-commodity relationships in Kenyan food commodity price data.
ii) To train supervised machine learning models for anomaly detection.
iii) To evaluate the scalability of the developed framework for application to similar market contexts.

1.4 Research Questions

The following research questions will be adopted to achieve the objectives of this study;

i) How can data engineering be designed to integrate seasonality, spatial variation and inter-commodity relationships in the Kenyan food commodity price data?
ii) How can supervised machine learning models be trained to detect anomalies in food commodity prices?
iii) What is the scalability of the developed ML framework for application in similar market contexts?

1.5 Significance and Justification of the Study

This study addresses an existing problem in Kenya's market monitoring capacity by developing a supervised machine learning framework capable of detecting anomalies in food commodity prices with accuracy and speed than current manual or threshold-based approaches (Jordan and Mitchell, 2015; Aggarwal, 2017). These improvements can enable policymakers, regulatory bodies and humanitarian agencies to respond effectively to real market disruptions, reducing the risk of prolonged instability in the food market. The study also advances academic understanding by demonstrating how tailored feature engineering, like capturing seasonality, spatial variation and inter-commodity relationships, can enhance model performance in agricultural price analysis (Newton et al., 2021). Most importantly, however, the framework design is meant to be architecturally scalable, which is critical to ensure adaptability with different datasets and market contexts with similar characteristics beyond Kenya (Mehendale, 2024).

1.6 Definition of Key Terms

The following key terms: food commodity and food prices are critical to the scope and methodology of this study. Therefore, providing a clear definition will help establish a shared understanding of both the technical and economic dimensions of this study, enabling proper alignment of the findings.

1.6.1 Food Commodity

For this study, a food commodity refers to raw agricultural products that are traded, processed or consumed in their primary form (Fontefrancesco, 2019). The examples here include staple cereals, legumes, root crops and unprocessed animal products, which form the foundation of national and global food systems.

1.6.2 Food Prices

Food prices, as defined by de Janvry (1982), are the monetary values assigned to agricultural and food commodities, which influence real incomes, wages and consumption patterns within a population. They are regarded as essential wage goods that shape employment, rural welfare and political stability. McCalla (2009) distinguishes between nominal prices, measured in current monetary terms and real prices, adjusted for inflation to reflect purchasing power.

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

This chapter reviews existing studies on the application of supervised machine learning techniques for detecting anomalies in food commodity price data. It examines how different algorithms have been used to identify irregular patterns, capture market dynamics and improve the accuracy of price monitoring. The review also highlights gaps in current approaches that the present study seeks to address,

2.2 Empirical Literature Review

2.2.1 Seasonality, Spatial and Inter-Commodity Feature Engineering in Food Prices

Food commodity markets are characterised by strong and predictable seasonal cycles tied to production, harvest and climate shocks. Empirical studies highlight the importance of accounting for these intra-year fluctuations using decomposition techniques like Seasonal-Trend decomposition using Loes (STL), Fourier terms and calendar dummies (Gilbert et al., 2017; Zhang et al., 2025). However, Gilbert et al. (2017) study on food price seasonality in Africa demonstrated that dummy-based seasonality often underestimates the amplitude of fluctuations, whereas the trigonometric and sawtooth techniques are more effective in capturing sharp trough-to-peak gaps that are commonly observed in African staple markets. In the context of Kenya, maize prices are a clear example which usually fluctuates by nearly one-third across harvest cycles.

Beyond cyclical dynamics, recent studies emphasise that agro-ecological signals substantially enhance the predictive capacity of food price models. Weather-based features such as rainfall anomalies, the Standardised Precipitation Index (SPI), and the Normalised Difference Vegetation Index (NDVI) have been widely integrated to capture production shocks and explain market volatility. A study by Shuaibu et al. (2016) found that NDVI anomalies strongly aligned with drought episodes and corresponded with significant maize price surges in Kenyan markets, underscoring the role of vegetation health as a leading indicator of price instability. Supportive evidence from Niger confirmed that reduced rainfall intensifies intra-seasonal price variability, often exerting a delayed effect on staple food availability and affordability.

Further, spatial heterogeneity, influenced by transport costs, cross-border trade, and infrastructural constraints, plays a critical role in shaping Kenya's food commodity price differentials. For example, Kihiu and Gachanja (2022) found that improved rural road infrastructure is associated with lower maize prices in Kenyan retail markets and that spatial price dependence exists across neighbouring counties affected by transport networks. Equally, spatial econometric tools like Moran's I and network-based methods have been used to test market integration and highlight the role of trade corridors in price transmission. For instance, Abdulai (2000), Waiswa (2023) and Waiswa and Yavuz (2023) study in Ghana and East Africa, respectively, employed cointegration tests and nonlinear ARDL models, which showed strong cross-border price transmission between Nairobi and Uganda markets, indicating that a 1% increase in maize prices in Nairobi leads to nearly a 0.9% increase in Ugandan prices. Such evidence illustrates that integrating spatial data, based on geographical distance, road connectivity, and neighbouring market price lags into the supervised machine learning models can bring meaningful improvement in detecting price anomalies.

Moreover, inter-commodity relations are critical in understanding the dynamics of commodity markets, as they often co-move under shared shocks. Kabundi and Zahid (2023) study highlighted the existence of global and group cycles in commodity markets that are largely driven by global supply shocks since 2000, which have increased the demand for commodities like grains and precious metals. Similarly, Esposti (2017)'s research supports the presence of a common latent factor that affects commodity price movements, with significant long-term equilibrium price increases observed in recent decades. However, regional studies revealed something different. In the Indian context, Kumar (2006) analysis revealed unique long-term relationships among certain cereals, pulses and oilseeds. Here, government policy instruments such as minimum support prices are provided as stabilisers against disturbances that might otherwise transmit freely into the commodity market. This raises questions as to whether the observed co-movements are truly global or are mediated by local policy frameworks. Other researchers have shifted the lens towards energy-food linkages. Mensi et al. (2014) highlights the dynamic spillovers between energy and cereal prices, where volatility in oil markets amplifies fluctuations in grain markets. Similarly, Kozian et al. (2025) tested methodological boundaries by comparing VAR and VARX models to machine learning approaches like Random Forests, ultimately finding that the econometric models capture cross-commodity dynamics more effectively.

2.2.2 Comparing Supervised ML Models against Threshold-based Monitoring

Studies on food price monitoring have tested supervised machine learning approaches against traditional threshold-based methods. Minot (2014) and Food and Agriculture Organization of the United Nations [FAO] (2025) studies demonstrate that rule-based z-scores and interquartile thresholds are often inadequate in volatile contexts, as they usually fail to capture regime shifts resulting from trade shocks and drought. This shortcoming has motivated the adoption of supervised machine learning methods that can better adapt to structural changes in price dynamics.

However, since labelled anomalies are rare in food market data, many studies have employed the use of synthetic anomaly generation. For instance, Ramírez et al. (2020) proposed a latent-space approach that distils inlier representations and synthesises pseudo-outliers from the outlier distribution, a method that enables robust classifier training even without notable examples of anomalies in data. Broadly, Jiang et al. (2023) reviewed weakly supervised anomaly detection techniques suitable for time-series and tabular data and presented techniques like incomplete, inexact and inaccurate supervision that align well with the challenge of food price anomaly detection labelling. Equally, researchers have proposed a combination of techniques like focal loss, which emphasises difficult samples over easy ones (Lin et al., 2018) and cost-sensitive learning, where loss function is weighted to disadvantage mistakes on minority classes (Baloch et al., 2019; Ahmad and Rahimi, 2025). As such, methods like Focused Anchor Loss exhibit the effectiveness of combining these approaches (Baloch et al., 2019), generally to cope with class imbalance that characterises the classification of anomalies.

A diverse set of literature demonstrates that supervised machine learning approaches like support vector machines, decision trees and gradient boosting outperform conventional threshold-based monitoring methods in anomaly detection. Minot (2014) criticises threshold methods for their inability to capture complex, nonlinear market dynamics and structural breaks. This is unlike the supervised machine learning models that leverage historical anomaly patterns and contextual features to identify both sudden shocks like drought-induced maize price spikes and slower, structural distortions tied to transport challenges or cross-border trade disruptions (Minot, 2014).

2.2.3 Scalability and Transferability of ML Model

A growing body of literature emphasises the importance of designing scalable models that adapt to new network settings and data environments.

Andrée and Pape (2023) study examined whether machine-learning models can fill gaps in market-level food price series across multiple fragile settings. Using partial survey data from over 1,200 markets in 25 countries, the approach trained commodity-specific models that impute missing prices from nearby markets and related items. In validation against held-out observations, the method captured about 85% of observed price variation even when 60% to 80% of survey data were missing, indicating a cross-country scalability for real-time monitoring.

However, the study noted that in fragile contexts where both price data and covaries are thin, it may not be possible to determine transferability due to instability in market linkages. The study, therefore, emphasised that while scalable, models still require a minimum density of observation to maintain accuracy.

Equally, Baquedano (2015) developed the FAO's Indicator of Food Price Anomalies (IFPA), which applied z-score thresholds to growth rates for anomaly detection. The method provided a standardised anomaly label that supports supervised machine learning in cross-country settings. The authors concluded that IFPA enhances transferability by harmonising anomaly definitions. However, Tadesse et al. (2014) criticism such statistical anomaly frameworks for overlooking structural weaknesses of volatility, such as cross-border trade policies or speculative dynamics, which weaken predictive robustness when applied in new contexts.

A study by the World Food Programme (2014) developed the ALPS (Alert for Price Spikes) indicator, which measures deviations from seasonal price expectations. ALPS has been praised for its operational utility, as it has been integrated into multiple country dashboards and reused to train supervised anomaly detectors. The findings concluded that ALPS is the best suited as a real-time trigger tool. However, critics aver that ALPS design is less suited to capturing gradual or sustained price increases that evolve over longer periods. A recent study by Busker et al. (2024) notes that while ALPS effectively identifies short-lived shocks, it tends to underperform when faced with slow-moving or cumulative price rises, thereby limiting its relevance for long-term policy planning and structural food market assessments. These findings suggest that ALPS is best applied as a complementary tool alongside broader analytical frameworks that track persistent market trends.

Further, Herteux et al. (2024) applied a reservoir-computing (RC) model to forecast food security trends in Mali, Nigeria, Syria and Yemen. The RC approach achieved a 5 percentage-point RMSE for 60-day forecasts and 0.43 accuracy on deterioration classification. Besides, the findings showed that the RC models performed better in forecasting food security trends than conventional methods like ARIMA and LSTM, particularly when extended beyond 5 forecast steps, maintaining lower error growth and offering more reliable predictions in deteriorating conditions. However, the ability of RC to identify rare but severe events remains limited, since these cases are underrepresented in the training data. Similar concerns are reflected in the findings of Busker et al. (2024), who noted that machine learning systems trained for broad cross-country applications often lose sensitivity to extreme shocks. These studies thus demonstrate that whereas RC enhances overall forecasting stability, it is still limited in detecting high-impact crises.

In addition, Foini et al. (2023) employed gradient-boosted models across six countries, achieving 99% one-day accuracy in Syria but dropping to 47% at 30 days in Yemen. The authors concluded that transferability is strongest for short horizons, but weak for longer-term warnings without retraining. Independent evaluations of early-warning systems support this limitation, noting that while such models achieve high overall accuracy, they often underperform in predicting the most severe crises, leading to missed or delayed alerts for high-impact events Maxwell et al. (2019). This suggests that while gradient-boosted approaches enhance short-term forecasting, their reliability in crisis-prone environments remains constrained.

2.2.4 Gaps in Related Works

A review of related works reveals three major emerging gaps. First, the existing works have not fully integrated climatic, spatial and inter-commodity features into supervised ML pipelines (Jordan and Mitchell, 2015; Minot, 2014; Aggarwal, 2017). This issue constrains the ability of the developed models to capture the complex drivers of food commodity price volatility fully. Second, anomaly labelling remains a challenge, often relying on synthetic data integration that risks deviating from real-world dynamics, thereby weakening the reliability of trained classifiers (Mehendale, 2024; Wang and Zhao, 2023). Third, the scalability and transferability of models remain uncertain, especially in fragile markets where rare but extreme shocks are dominant (Aggarwal, 2017; Newton et al., 2021). As a result, this study addresses the underlying gaps by developing supervised ML models that incorporate richer feature sets, apply advanced methods for handling class imbalance and test scalability in Kenyan food commodity markets with potential for regional application.

CHAPTER 3: METHODOLOGY

This study applied the Cross-Industry Standard Process for Data Mining (CRISP-DM) as the methodological framework. CRISP-DM is widely recognised in applied data science and offers a structured yet iterative process that links problem definition to deployment (Wirth and Hipp, 2000; Durango et al., 2023). Its modular design is particularly relevant for anomaly detection in food commodity markets, where heterogeneous datasets and structural shocks complicate analysis (Food and Agriculture Organization of the United Nations [FAO], 2025). The stages followed in this study were data understanding, collection, cleaning, model development, performance evaluation, and deployment, as shown in Figure 3.1.

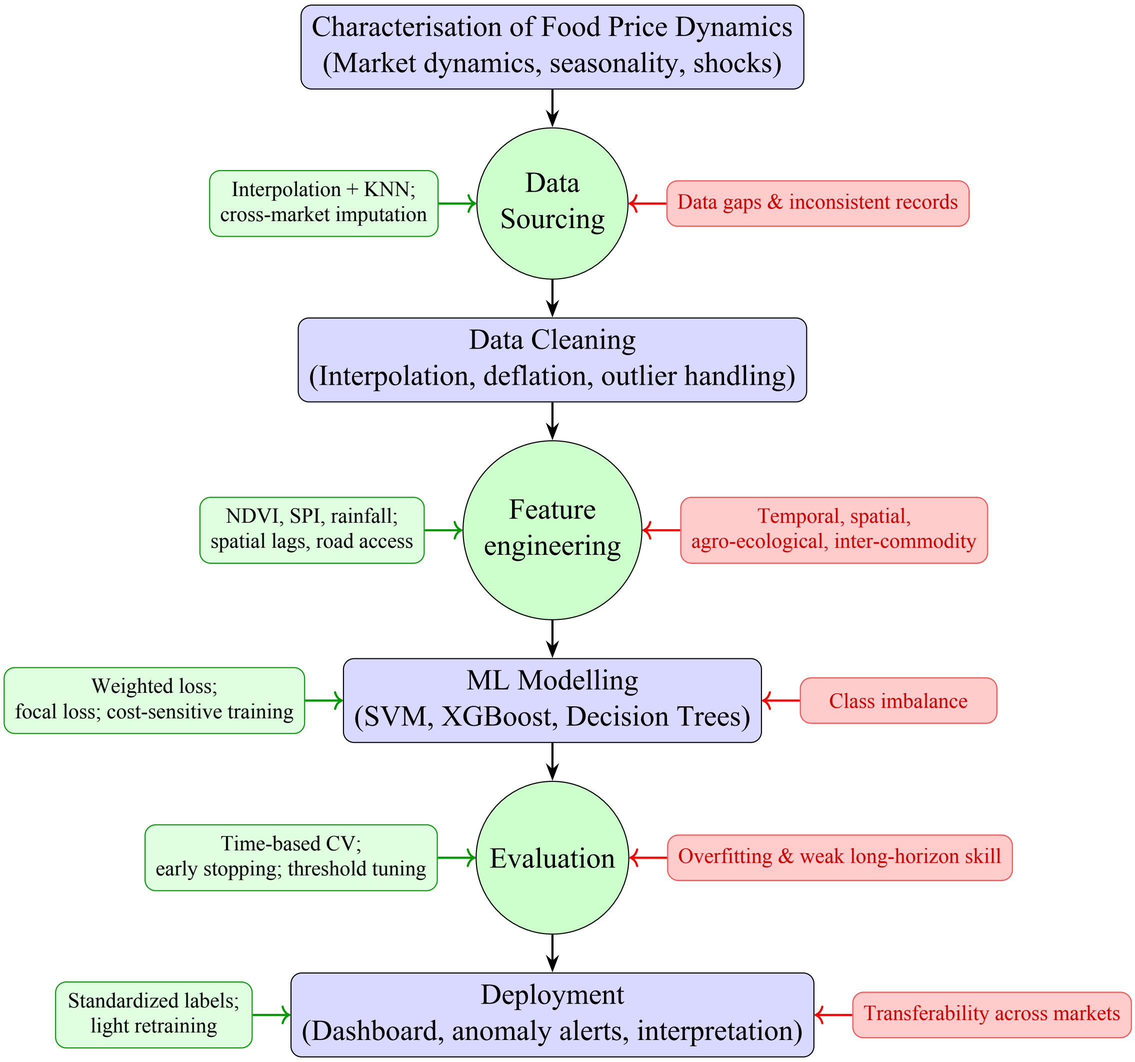

Figure 3.1: Methodology workflow with stages (rectangles), checkpoints (circles), challenges (red), and solutions (green) for supervised ML anomaly detection in food commodity prices.

3.1 Characterisation of Food Price Dynamics

In this study, the characterisation stage involved establishing the structural properties of food commodity price series in Kenya by analysing their patterns, spatial dependencies and exposure to irregular shocks. Historical data on staples such as maize, beans and rice were examined for recurrent harvest-linked cycles using decomposing techniques to isolate trend and seasonal components, as recommended in previous studies on African food price volatility (Gilbert et al., 2017; Minot, 2014). Further, spatial linkages were assessed by incorporating market connectivity indicators, such as distance to Nairobi and proximity to cross-border trade routes, which have been shown to significantly influence price transmission across counties and neighbouring countries (Kihiu and Gachanja, 2022; Waiswa, 2023). In addition, climate variability was captured using rainfall anomalies and NDVI indices, reflecting the evidence that agro-ecological signals strongly predict drought-induced price surges (Shuaibu et al., 2016). This characterisation provided the basis for defining anomalies not as random outliers, but as deviations from expected price trajectories once seasonal, spatial and climatic dynamics are accounted for. This was intended to ensure that the subsequent modelling is both data-driven and context-sensitive.

3.1.1 Data Sourcing

The dataset for this study was primarily collected from the World Food Programme (WFP) Price Database via the OCHA Services platform for the period of two hundred and thirty-eight (238) months, starting 15 January 2006 to 15 November 2025. This source provided monthly retail and wholesale prices of major staples, including maize, beans and rice, across different Kenyan markets. The primary data was complemented with official consumer price index and market-level retail statistics data sourced from the Kenya National Bureau of Statistics (KNBS) to ensure consistency with national economic indicators. The FAO Food Price Monitoring and Analysis (FPMA) tool provided a sample of international reference prices and comparative regional data to facilitate checks and contextualization of domestic price movement. At the same time, FEWS NET market bulletins was used to extract early warning price data and qualitative assessments of shocks such as droughts, trade restrictions and supply chain disruptions.

Further, the study incorporated auxiliary datasets to capture non-price volatility drivers. These included rainfall anomalies and the SPI for climatic variability, the NDVI for crop and pasture conditions and infrastructure indicators such as road density, distance to Nairobi and proximity to trade corridors for spatial dependencies. The integration of WFP market data with official statistics, international reference series and agro-ecological variables provided broad temporal coverage and contextual depth, reflecting recommendations from previous work on food price modelling (Minot, 2014; Shuaibu et al., 2016). Table 1 summarises the data sources and type of information that the study obtained and utilised in the implementation of this project.

Table 1: Data sources and type of information collected for the study

Source	Type of Data Collected	Purpose/Use in Study
World Food Programme (WFP) Price Database via OCHA Services	Monthly retail and wholesale food prices for major staples (maize, beans, rice) across Kenyan markets	Main dataset for anomaly detection; provided consistent, harmonised price series across regions
Kenya National Bureau of Statistics (KNBS)	Consumer price indices (CPI) and official retail market prices	Adjusted for inflation and aligned local price series with national economic indicators
FAO Food Price Monitoring and Analysis (FPMA) Tool	International and regional reference food prices	Enabled comparison of Kenyan prices with global and regional benchmarks; supported cross-checks
FEWS NET Market Bulletins	Early warning market prices and qualitative assessments of shocks (e.g., droughts, trade bans, supply chain disruptions)	Provided context on external shocks and qualitative explanations for abnormal price behaviour
Auxiliary Datasets (Climate and Infrastructure)	Rainfall anomalies, Standardised Precipitation Index (SPI), Normalised Difference Vegetation Index (NDVI), road density, distance to trade hubs	Captures agro-ecological and spatial factors affecting price dynamics and transmission

3.1.2 Data Cleaning and Preprocessing

The collected datasets were subjected to preprocessing to ensure they are suitable for a supervised ML model for anomaly detection. The missing observations in short intervals were treated using linear interpolation, while longer gaps were addressed through K-nearest neighbour (KNN) imputation that leverages correlations between neighbouring markets, as applied in recent market integration studies by Waiswa (2023) and Waiswa and Yavuz (2023). Outlier screening was performed to remove clear reporting errors, but genuine spikes were retained since they represent anomalies of interest. All prices were converted into real terms using the CPI and log-transformation to stabilise variance. These steps ensured that the resulting dataset is clean and harmonised while preserving the underlying economic signals necessary for the development of a robust anomaly detection.

3.1.3 Exploratory Data Analysis

The exploratory data analysis (EDA) was undertaken to reveal the structural properties of the Kenyan food price series and assess their suitability for supervised anomaly detection. Initial exploration involved visual inspection using time-series plots, decomposition charts and heatmaps to detect seasonal cycles, long-term trends, and sudden disruptions, in line with established approaches to commodity market analysis (Gilbert et al., 2017; Minot, 2014). Growth rates were computed using the formula in Equation 1 to quantify short-term price movements:

$$r_t = \frac{P_t - P_{t-1}}{P_{t-1}} \times 100, \tag{1}$$

where r_t denotes the period-to-period price change. Volatility was then captured using rolling standard deviations of returns as shown in Equation **??**:

$$\sigma_t = \sqrt{\frac{1}{k}\sum_{i=0}^{k-1}(r_{t-i} - \bar{r})^2}, \tag{2}$$

with k representing the chosen rolling window length, allowing detection of instability during shocks such as droughts or trade restrictions. Subsequently, spatial and agro-ecological linkages were examined through correlation analysis, expressed as shown in Equation 3:

$$\rho_{xy} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}, \tag{3}$$

to measure the co-movement between market prices and explanatory variables such as rainfall anomalies of NDVI (Shuaibu et al., 2016). These diagnostic steps enabled the identification of candidate anomalies, highlighting the influence of seasonal and climatic drivers.

3.2 Machine Learning Modelling

The modelling stage constituted the main component for this study, where supervised ML techniques were proposed for detecting anomalies in food commodity prices. The modelling process workflow began with feature engineering before proceeding to model design and training, and culminating in the classification of anomalous behaviour.

3.2.1 Feature engineering

In this study, feature engineering was carried out to construct explanatory variables that capture the complex drivers of food price behaviour. The process was based on four dimensions of influence.

First, the temporal features were generated through seasonal dummies, Fourier terms and lagged prices to represent harvest cycles and cyclical variation. Secondly, spatial features were introduced using neighbouring market prices, distance to Nairobi and road accessibility indicators to reflect trade networks and regional dependencies. Thirdly, agro-ecological features were derived from rainfall anomalies, the SPI and NDVI, providing signals of production shocks and drought stress. Finally, the inter-commodity features were created through price ratios and co-movement indices between staple crops and energy markets to capture cross-market linkages. These features were meant to enrich the input space of the machine learning models, ensuring that anomaly detection reflects both statistical deviations and the underlying economic and ecological realities of Kenyan food markets.

3.2.2 Model Selection

The supervised anomaly detection task was formulated as a binary classification problem. This means that, for each observation x_i, the objective is to predict a label $y_i \in \{0, 1\}$, with 1 denoting an anomaly and 0 denoting a normal observation. The central objective is to approximate a function $f : X \to Y$ that minimises a weighted loss function, $L(y, \hat{y})$ designed to handle the class imbalance inherent in anomaly detection

In this study, the model selection was guided by the need to balance predictive performance, interpretability and robustness in detecting food price anomalies. Three supervised machine learning algorithms were considered, that is, the One Class Support Vector Machines (SVM), Extreme Gradient Boosting (XGBoost) and Decision Trees.

i) *Support Vector Machines (SVM)*

The SVMs were preferred for their suitability in anomaly detection because they construct decision boundaries that separate normal and anomalous observations, even in high-dimensional spaces (Erfani et al., 2016). These models classify anomalies by finding a hyperplane that maximises the margin between normal and anomalous classes. The decision function is expressed as:

$$f(x) = \text{sign}\left(\sum_{i=1}^{n} \alpha_i y_i K(x_i, x) + b\right), \tag{4}$$

where α_i are the support vector weights, y_i are the class labels, $K(\cdot, \cdot)$ is the kernel function, and b is the bias term (Cortes and Vapnik, 1995). This formulation allowed SVMs to handle non-linear relationships common in food price data.

ii) Gradient Boosting (XGBoost)

The XGBoost was preferred for modelling complex non-linear interactions among temporal, spatial and agro-ecological features more effectively than simpler classifiers. Its scalability and efficiency also make it well-suited for handling multi-market food price data, where anomalies are rare (Chen and Guestrin, 2016). The optimisation objective combines a regularised loss and penalty on model complexity as expressed in Equation 5 and Equation 6, respectively:

$$\mathcal{L}(\phi) = \sum_{i=1}^{n} l\left(y_i, \hat{y}i\right) + \sum k = 1^K \Omega(f_k), \tag{5}$$

with

$$\Omega(f) = \gamma T + \tfrac{1}{2}\lambda \|w\|^2, \tag{6}$$

where l is the differentiable loss function, such as logistic loss, f_k are individual trees, T is the number of leaves, γ controls tree complexity, λ regularises leaf weights, and w represents leaf weights. This ensures a balance between model flexibility and overfitting control.

iii) Decision Trees

Decision Trees was included as a baseline model because of their interpretability and ability to capture non-linear relationships in food price dynamics. At each node, the tree selects a feature split that maximises the information gain by reducing impurity in the child nodes. Impurity is often measured using entropy expressed in Equation 7:

$$H(S) = -\sum_{c=1}^{C} p_c \log_2(p_c), \tag{7}$$

where p_c is the proportion of observations in class c, and C is the number of classes.

Equation 8 expresses that the best split is chosen to maximise the information gain:

$$IG(S, A) = H(S) - \sum_{v \in \text{Values}(A)} \frac{|S_v|}{|S|} H(S_v), \tag{8}$$

where $IG(S, A)$ is the information gain of splitting dataset S on attribute A, and S_v is the subset of S corresponding to value v.

iv) Weighted Cross-Entropy Loss

Since anomalies are rare, the loss function was adjusted to give higher weight to misclassified anomalies as illustrated in Equation 9:

$$L(y, \hat{y}) = -w_1\, y \log(\hat{y}) - w_0\, (1 - y) \log(1 - \hat{y}), \tag{9}$$

where $w_1 > w_0$ ensures that anomalies contribute more strongly to the optimisation objective.

3.2.3 Hyperparameter Tuning

This study conducted systematic hyperparameter tuning to achieve optimal model performance. For the SVM, the penalty parameter C, kernel choice, and kernel-specific settings such as the RBF with γ was adjusted to manage the balance between margin size and classification errors. For the XGBoost, parameters including the learning rate η, maximum depth, subsampling ratio and the regularisation terms γ and λ were calibrated to reduce overfitting while maintaining predictive strength. In the case of Decision Trees, tuning focused on maximum depth, minimum samples required for a split and the splitting criterion (entropy or Gini), to improve accuracy without losing interpretability.

3.3 Performance Evaluation

The model's performance was evaluated using a stratified 5-fold cross-validation design to test robustness across time and markets. The main metrics were defined as shown in Table 2.

Table 2: Performance evaluation metrics used in the study

Metric	**Description**	**Mathematical Definition**
Precision	Correctly identified anomalies among all predicted anomalies	$\text{Precision} = \frac{TP}{TP+FP}$
Recall	Actual anomalies correctly identified	$\text{Recall} = \frac{TP}{TP+FN}$
F1-score	Harmonic mean of precision and recall	$F1 = 2 \times \frac{\text{Precision}\times\text{Recall}}{\text{Precision}+\text{Recall}}$
True Positive Rate	Proportion of anomalies detected	$TPR = \frac{TP}{TP+FN}$
False Positive Rate	Proportion of normal cases incorrectly flagged	$FPR = \frac{FP}{FP+TN}$
Area Under the Curve	Overall ability of the model to separate normal from anomalous cases	$\text{AUC} = \int_0^1 \text{TPR}(\text{FPR})\, d(\text{FPR})$

Lastly, the confusion-matrix analysis complemented these metrics to show trade-offs between false alarms and missed anomalies. Because missed anomalies carry greater policy risk than false positives, recall received greater weight in the final model selection.

3.4 Model Deployment

Figure 3.2 is an illustration of the deployment process for the final model, starting with data input to and market use.

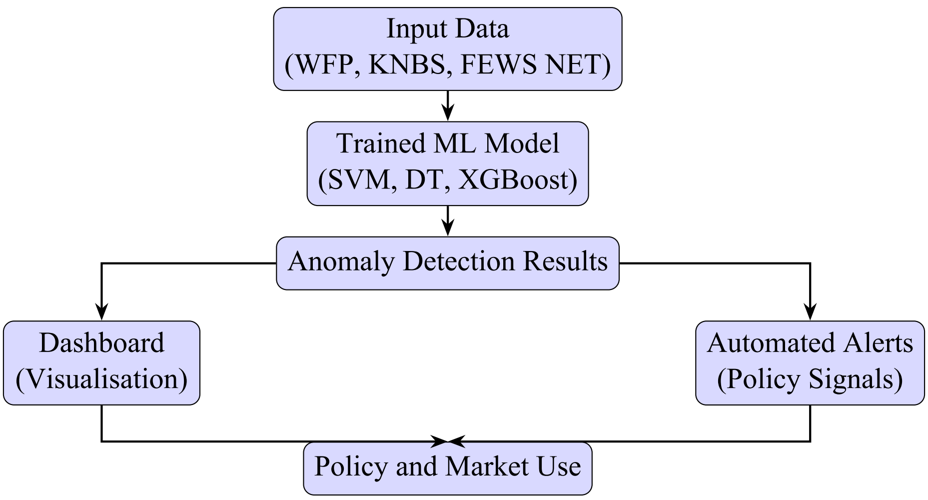

Figure 3.2: Deployment workflow for anomaly detection outputs integrated into dashboards and alerts.

The model was deployed as a cloud-based system that connects directly to market data from WFP, KNBS, and FEWS NET. New data was processed in real time by the trained algorithms to detect anomalies and produce price forecasts. Results feed into two tools: a dashboard that visualises market patterns and anomalies against historical trends, and an alert system that issues early warnings on unusual changes in staple food prices such as maize, beans, and rice. These outputs supports farmers, traders, and policymakers by providing timely insights for both immediate response and long-term planning. The use of cloud infrastructure was meant ensure fast processing, secure access, and scalability, while automated monitoring and continuous learning keep the models updated and adaptable to changing market and climate conditions.

3.5 Expected Outcomes

This study provide a supervised machine learning framework that identifies food price anomalies in near real time using routinely collected market data. The model brings seasonality, spatial differences across Kenyan markets and links between commodities into one pipeline to lift detection accuracy over standard baselines. The workflow was packaged so it can be ported to additional markets with minimal reconfiguration, allowing replication in neighbouring countries once local data is available.

The practical output was intendeto be decision-ready signals for analysts and policymakers. The system generates weekly anomaly alerts, summary briefs and simple visual explanations that show why a flag was raised and how current prices compare with seasonal norms. These outputs are meant to strengthen routine market monitoring, support targeted policy actions such as temporary subsidies or import timing and inform humanitarian planning before sharp price swings spill over into food insecurity. By shortening the time from data ingestion to a clear alert, the project aims to improve early warning agencies that track market stability.

CHAPTER 4: SYSTEM ARCHITECTURE AND DESIGN

CHAPTER 5: SYSTEM DEVELOPMENT, TESTING AND VALIDATION

CHAPTER 6: CONCLUSIONS, RECOMMENDATIONS AND FUTURE WORKS

Bibliography

Abdulai, A. (2000). Spatial price transmission and asymmetry in the ghanaian maize market. *Journal of Development Economics*, 63(2):327–349.

Aggarwal, C. C. (2017). *Outlier Analysis*. Springer, Cham, 2 edition.

Ahmad, H. M. and Rahimi, A. (2025). Dynamic focal loss for imbalanced learning. *SN Computer Science*, 6:657. Published 16 July 2025.

Andrée, B. P. J. and Pape, U. J. (2023). Machine learning imputation of high frequency price surveys in papua new guinea. Policy Research Working Paper 10559, World Bank, Washington, DC.

Awange, J. L., Aluoch, J., Ogallo, L. A., Omulo, M., and Omondi, P. (2007). Frequency and severity of drought in the lake victoria region (kenya) and its effects on food security. *Climate research*, 33(2):135–142.

Baloch, B. K., Kumar, S., Haresh, S., Rehman, A., and Syed, T. (2019). Focused anchors loss: cost-sensitive learning of discriminative features for imbalanced classification. In Lee, W. S. and Suzuki, T., editors, *Proceedings of the Eleventh Asian Conference on Machine Learning*, volume 101 of *Proceedings of Machine Learning Research*, pages 822–835. PMLR.

Baquedano, F. G. (2015). Developing an indicator of price anomalies as an early warning tool: A compound growth approach. Technical report, Food and Agriculture Organization of the United Nations, Rome.

Brown, M. E. and Kshirsagar, V. (2015). Weather and international price shocks on food prices in the developing world. *Global Environmental Change*, 35:31–40.

Busker, T., van den Hurk, B., de Moel, H., van den Homberg, M., van Straaten, C., Odongo, R. A., and Aerts, J. C. J. H. (2024). Predicting food-security crises in the horn of africa using machine learning. *Earth's Future*, 12(8):1–20.

Chen, T. and Guestrin, C. (2016). Xgboost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, KDD '16, pages 785–794, New York, NY, USA. Association for Computing Machinery.

Cortes, C. and Vapnik, V. (1995). Support-vector networks. *Machine Learning*, 20(3):273–297.

de Janvry, A. (1982). Why do governments do what they do? the case of food price policy. Working paper 198265, CUDARE Working Paper Series, University of California, Berkeley.

Durango, Vanegas, C. E., Mejía, J. C. G., Vargas Agudelo, F. A., and Soto Durán, D. E. (2023). A representation based on essence for the crisp-dm methodology. *Computación y Sistemas*, 27(3):—.

Erfani, S. M., Rajasegarar, S., Karunasekera, S., and Leckie, C. (2016). High-dimensional and large-scale anomaly detection using a linear one-class svm with deep learning. *Pattern Recognition*, 58:121–134.

Esposti, R. (2017). What makes commodity prices move together? an answer from a dynamic factor model. Conference paper, Research Papers in Economics. Presented at the 2017 European Association of Agricultural Economists (EAAE) Congress.

FAO, IFAD, UNICEF, WFP and WHO (2022). The state of food security and nutrition in the world 2022: Repurposing food and agricultural policies to make healthy diets more affordable. Report, Food and Agriculture Organization of the United Nations, Rome.

Foini, P., Tizzoni, M., Martini, G., Paolotti, D., and Omodei, E. (2023). On the forecastability of food insecurity. *Scientific Reports*, 13(1):2793.

Fontefrancesco, M. F. (2019). Food commodity market: History and impact of food trading toward sdg2. In Leal Filho, W., Azul, A., Brandli, L., Özuyar, P., and Wall, T., editors, *Zero Hunger*, Encyclopedia of the UN Sustainable Development Goals. Springer, Cham. First Online: 24 July 2019.

Food and Agriculture Organization of the United Nations [FAO] (2025). Food price monitoring and analysis (fpma). https://www.fao.org/prices/home/food-price-monitoring. Accessed: 2025-08-17.

Gbadegesin, T. K., Andrée, B. P. J., and Braimoh, A. K. (2024). Climate shocks and their effects on food security, prices, and agricultural wages in afghanistan. Technical Report 10999, World Bank.

Gbashi, S. and Njobeh, P. B. (2024). 1. enhancing food integrity through artificial intelligence and machine learning: A comprehensive review. *Applied Sciences*.

Gilbert, C. L., Christiaensen, L., and Kaminski, J. (2017). Food price seasonality in africa: Measurement and extent. *Food Policy*, 67:119–132.

Gustafson, D. J. (2013). Rising food costs and global food security: Key issues and relevance for india. *The Indian Journal of Medical Research*, 138(3):398–410.

Herteux, J., Räth, C., Martini, G., Baha, A., Koupparis, K., Lauzana, I., and Piovani, D. (2024). Forecasting trends in food security with real time data. *Communications Earth & Environment*, 5:611.

Ilaha, A. (2022). 4. research methods of food marketing. *International Journal of Innovative Technologies in Economy*.

Jiang, M., Hou, C., Zheng, A., Hu, X., Han, S., Huang, H., He, X., Yu, P. S., and Zhao, Y. (2023). Weakly supervised anomaly detection: A survey. *arXiv preprint arXiv:2302.04549*. Code available at: https://github.com/Minqi824/WSAD-Survey.

Jordan, M. I. and Mitchell, T. M. (2015). Machine learning: Trends, perspectives, and prospects. *Science*, 349(6245):255–260.

Kabundi, A. and Zahid, H. (2023). Commodity price cycles: Commonalities, heterogeneities, and drivers. Policy Research Working Paper 10401, World Bank.

Kihiu, E. N. and Gachanja, J. N. (2022). Discussion Paper No 293 of 2022 on Access to Maize Retail Markets and the Impacts of Rural Road Infrastructure in Kenya. Discussion Paper 293, Kenya Institute for Public Policy Research and Analysis (KIPPRA).

Korir, L., Rizov, M., and Ruto, E. (2020). 7. food security in kenya: Insights from a household food demand model. *Economic Modelling*.

Kozian, L. L., Machado, M. R., and Osterrieder, J. R. (2025). Modeling commodity price co-movement: Building on traditional time series models and exploring applications of machine learning algorithms. *Finance and Stochastics*. Online First.

Kumar, P. (2006). Inter commodity price linkages in india: A case of foodgrains, oilseeds and edible oils. *The Journal of International and Area Studies*, 13:83–100.

Lin, T.-Y., Goyal, P., Girshick, R., He, K., and Dollár, P. (2018). Focal loss for dense object detection. *arXiv preprint arXiv:1708.02002*. Version 2, last revised 7 Feb 2018. Code available online.

Maxwell, D., Hailey, P., Spainhour Baker, L., and Kim, J. J. (2019). Constraints and complexities of information and analysis in humanitarian emergencies: Evidence from yemen. Technical report, Feinstein International Center, Tufts University, Boston, MA.

McCalla, A. F. (2009). World food prices: Causes and consequences. *Canadian Journal of Agricultural Economics*, 57(1).

Mehendale, P. (2024). Scalable architecture for machine learning applications. *Journal of Scientific and Engineering Research*, 11(8):111–117.

Mensi, W., Hammoudeh, S., Nguyen, D. K., and Yoon, S.-M. (2014). Dynamic spillovers among major energy and cereal commodity prices. Working Paper 2014-160, IPAG Business School.

Minot, N. (2014). Food price volatility in sub-saharan africa: Has it really increased? *Food Policy*, 45:45–56.

Newton, C., Singleton, J., Copland, C., Kitchen, S., and Hudack, J. (2021). Scalability in modeling and simulation systems for multi-agent, ai, and machine learning applications. In *Artificial Intelligence and Machine Learning for Multi-Domain Operations Applications III*, volume 11746, pages 534–552. SPIE.

Nyoro, J. K., Kirimi, L., and Jayne, T. S. (2004). Competitiveness of kenyan and ugandan maize production: Challenges for the future. Working Paper 10, Tegemeo Institute of Agricultural Policy and Development, Egerton University.

Pitchaimuthu, V. and Holla, A. (2024). 2. machine learning technique for anomaly detection. *International journal of engineering technology and management sciences*.

Ramírez, A., Khan, A., Bekkouch, I. E. I., and Sheikh, T. S. (2020). Anomaly detection based on zero-shot outlier synthesis and hierarchical feature distillation. *arXiv preprint arXiv:2010.05119*. To appear in IEEE Transactions on Neural Networks and Learning Systems.

Santoro, D., Ciano, T., and Massimiliano Ferrara (2024). 1. a comparison between machine and deep learning models on high stationarity data. *Dental science reports*.

Shuaibu, S., Ogbodo, J., Wasige, J., and Mashi, S. (2016). Assessing the impact of agricultural drought on maize prices in kenya with the approach of the spot-vegetation ndvi remote sensing. *Journal of Geoscience and Environment Protection*, 4(12):8–18.

Tadesse, G., Algieri, B., Kalkuhl, M., and von Braun, J. (2014). Drivers and triggers of international food price spikes and volatility. *Food Policy*, 47:117–128.

Waiswa, D. (2023). Spatial price transmission of maize grain prices among markets in kenya, tanzania, and uganda: Evidence from the nonlinear ardl model. *East African Scholars Journal of Economics, Business and Management*, 6(2):43–52.

Waiswa, D. and Yavuz, F. (2023). Market integration and asymmetric price transmission in selected domestic markets for major staple foods in uganda. *Future Business Journal*, 9(1):1–26.

Wang, L. and Zhao, X. (2023). Anomaly detection in time series data using improved methods. *Applied Sciences*, 13(10):6353.

Wirth, R. and Hipp, J. (2000). CRISP-DM: Towards a standard process model for data mining. In *Practical Applications of Knowledge Discovery and Data Mining*, pages 29–39, Berlin, Heidelberg. Springer.

World Food Programme (2014). Calculation and use of the alert for price spikes (alps) indicator: Technical guidance note. Technical report, World Food Programme, Rome. Technical Guidance Note, Analysis and Nutrition Service.

Yusuf, F. A. and Adelabu, S. O. (2024). Sowing prosperity: Analyzing the linkages between food prices and conomic well-being in sub-saharan africa. *Top Academic Journal of Economics and Statistics*, 9(1):32–45.

Zhang, L., Wang, F., Wang, K., He, Z., Chen, C., Liu, J., Wang, C., and Wang, Z. (2025). Improving agricultural commodity allocation and market regulation: a novel hybrid model based on dual decomposition and enhanced bilstm for price prediction. *Frontiers in Sustainable Food Systems*, 9.

Appendices